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In this video (13:25 minutes), Matt Woods and Dr Philip Ireland discuss vegetation carbon projects and how the ACCU Scheme works.

MATT WOODS: Hello. I'm Matt Woods, and with me is doctor Philip Ireland. Phil has worked in climate change policy for 20 years. And is currently the CEO of Carbon Neutral, one of Australia's oldest carbon project developers. He's worked on large scale environmental initiatives at a global level, is a founder and director of Hone Carbon, a soil carbon measurement startup, and a director of the national Soil Carbon Industry Group. Phil has a PhD in Climate Change adaptation and has worked in UN climate change negotiations. Phil, why is carbon farming as a mitigation to climate change important?

DOCTOR PHILIP IRELAND: Thank you. Carbon farming is incredibly important. In my work in climate change over 20 years, most of what I've done has been working on policy to reduce emissions going into the atmosphere. But as my career has progressed, I've recognised and the broader scientific community has recognised that we need to also draw carbon emissions out of the atmosphere if we're to achieve any of our targets and keep a safe climate. One of the best ways to draw carbon out of the atmosphere is through agricultural methods, through sequestering carbon into biomass, but also into soil organic carbon.

WOODS: In Australia, agriculture accounts for about seventeen percent of our emissions, can agriculture be seen as part of solution to climate change?

IRELAND: Agriculture has to be part of the solution. We can't achieve our targets without it.

One of the really interesting facts that has come out recently is that if we were to take all of the world's agricultural soils, and increase soil organic carbon by one percent in one metre of those soils, we would draw almost half of the historical emissions of our species out of the atmosphere. Soils and biomass have a huge potential to help us get to our net zero targets.

WOODS: I'm gonna move to carbon credits now. How does that system work in Australia, the carbon credit system?

IRELAND: So the Australian carbon credit system is one of the longest-standing carbon credit systems in the world, and it's been developed over successive governments. There are a number of methods that we use. In very broad terms, the system works by incentivising individuals and businesses to reduce their emissions, to avoid potential emissions and to draw carbon dioxide out of the atmosphere.

Using methods that sit under those broad three categories, individuals and businesses are then awarded a carbon credit unit or in Australia, we call it an ACCU, an Australian Carbon Credit Unit. That unit represents a tonne of carbon and that, once it's generated, can be sold and traded.

WOODS: Okay. I think there's a bit of confusion with farmers and landowners about how mature this market is. Is this something they can do today? Can they get involved in it now?

IRELAND: Absolutely. The market was first established a bit over ten years ago around 2011, and since it has had a number of changes and evolutions.

It's become a lot more sophisticated in recent years, with a number of very specific methodologies for very specific approaches to reducing emissions and drawing carbon out of the atmosphere. On the ACCU scheme website, the methods for farmers and landholders fall under the sections of agriculture and vegetation.

Under agriculture, there are a number of methods to reduce animal waste and effluent going into the atmosphere. There's also a method for increasing soil organic carbon, which is one of the fastest growing methods, and I believe one of the most exciting.

Under vegetation, there are a number of different things that landholders can do. They can plant plantations that have to remain in the ground for a set number of years, and they can also do reforestation using native and biodiverse species.

So carbon farming has a whole range of benefits for landholders, and a lot of these aren't talked about. Upfront, better management of your land for carbon can increase yield and profitability of your property. We know, greater soil organic carbon can reduce the need for more expensive fertilisers and chemicals.

We can think about carbon also as another crop. You're going in a process of de-risking the revenue of your property by providing more possible pathways for revenue. So that's number one, increasing yields and profitability.

Number two is climate change adaptation and resilience.

You're increasing your farm's and your land's ability to adapt to changes that are happening and are coming into the climate. We know increased soil organic carbon can increase water retention and water holding capacity of soils for times of drought. We know that trees and windbreaks can protect livestock from inclement weather, and we know the weather may become more windy and more unpredictable into the future. So you're making your property more resilient and adapted.

The third thing is that you're also contributing to climate change mitigation.

Farmers and landholders can get involved right now. They can register projects. They can be drawing carbon out of the atmosphere, and they can be generating extra revenue for their properties.

WOODS: Great. I'm sure every farmer's interested in extra revenue. What about the risks, though, to their business?

IRELAND: There are a number of risks. There are risks in all aspects of agriculture.

I'd encourage anyone looking at getting involved in carbon farming to do good due diligence on the people that they're working with and the methods.

The key thing to keep in mind is the long term obligations and commitments that you're putting your farm and future generations that are managing your farm under.

Under most carbon credit methods, you're making a commitment to permanence. We call it a permanence period of 25 to a hundred years. So in the instance of soil carbon, you're committing to draw carbon out of the atmosphere and then hold that carbon in the soil for a period of time. In vegetation and biomass methods, you're committing to growing trees, plants and shrubs and keeping those trees, plants and shrubs growing, and on the land, for a set period of time. There are obviously other risks associated with it as well, which are covered in this course.

One is that there is some volatility in the market for carbon as there are in all agricultural commodities. You can think about carbon as another crop. Another thing that you're growing on your property and you have to manage that in a similar way you have to manage other commodities that you're producing.

WOODS: Okay. So if I'm a farmer or land manager, and I want to get involved in it today, what's your top line advice of where to start and what to do?

IRELAND: There are lots of carbon project developers out there. They'll have different revenue and partnership models, which will suit different farmers in different ways. Definitely look into what they're offering and how they're getting paid for that offering. Is it an upfront fee or is it a long term share of the carbon credits? Both are legitimate ways, but they'll have different applications for you and different prices.

The second thing I would say is whilst it can be overwhelming, don't be overwhelmed and take your time. This isn't going anywhere, And over time, there will become more requirements, not just of farmers and landholders, but of all businesses - to be measuring their greenhouse gas emissions, reducing where they can, and in some cases, like landholders, drawing carbon out of the atmosphere. And engaging with this system will ultimately hopefully help for your farm's productivity and resilience to a changing climate.

So the first place to start is with the government website. So there's this thing called the ACCU scheme, the A-C-C-U scheme. It used to be called the Emissions Reduction Fund. You might be familiar with that. And this all sits under a government agency called the Clean Energy Regulator.

Over many years, they've made those materials more accessible, but that's your first stop when dealing with anything like this, because those websites cover the policy. They also are the doorways into registering projects and all these types of things.

From there, there are lots of carbon project developers and advisors that sit around that. But all of those bodies as well will be referring back to the government website.

WOODS: Okay. Any advice for people looking into this about potential consultants that they might want to get involved with?

IRELAND: There are a number of different consultants. I obviously work for one of those companies. There are ten to fifteen companies in Australia; a simple web search will bring up a number of these.

My encouragement would be to: If you want to bring on an advisor or a carbon project developer, look around, do your due diligence, really try to understand what their revenue model is. Some carbon

project developers will support you for an upfront fee, and that's all. Other carbon project developers may not charge you a fee, but will take a share of your long term carbon.

Other carbon project developers will partner with you and help you generate the carbon, whereas others will stand back. So depending on what your needs are, there will be an advisor and a developer to help you, but definitely look around.

WOODS: Maybe ask some other people that have been involved, that you trust; farmers, land managers?

IRELAND: Absolutely. You should be able to find some other farmers in your region who have done something in this space. And if not, there are networks on social media and other places. There are also more and more media articles about this. Landline has started covering this. It's on the ABC. So you should be able to verify who's around and what they're doing.

WOODS: Terrific. Final question, Phil. What is the win-win-win scenario for carbon farming?

IRELAND: So I believe for farmers, there are at least three wins with engaging carbon farming. It's increased farm profitability, it's farm adaptation to climate change and it's contribution to mitigation of climate change. On farm profitability, managing and increasing carbon on your property can increase yields of crops.

For example, we know that increased soil organic carbon increases the fertility of the soil, which can reduce the need for more expensive, and also carbon intensive fertilisers.

Windbreaks can help sheep and other animals survive inclement weather. And carbon could be thought of as another crop in your property. So when you're growing a crop, you've got your above-ground yield with whatever you're producing. You've got your below-ground yield of carbon. So it's contributing to de-risking of your future revenue flows by finding more diversified income bases.

Secondly, with adaptation, we know we're facing currently a changing climate, and it will change further into the future. A healthy farm, a farm with greater soil organic carbon with more trees, is a farm that's more resilient to those changes. We know soil with more soil organic carbon can hold and retain more water for dry times. We know trees form windbreaks and shade for animals, and there are a whole range of other co-benefits.

And finally, with mitigation, I believe over the long term, most farms will be drawing more carbon out of the atmosphere than they're actually putting into the atmosphere.

So farms and landholders are gonna play huge role in the mitigation of climate change, which isn't just good for future generations and the environment. It's going to be good for profitability of that property.

WOODS: Terrific. Thanks, Phil. I really appreciate your time.

IRELAND: My pleasure. Thanks for having me.

The ACCU Scheme

The [ACCU Scheme](#) offers farmers and land managers opportunities to run projects in Australia that reduce or avoid GHG emissions or store more carbon. Carbon farming projects, as well as energy efficiency, waste, transport and industrial processes projects, can take part in the scheme. The scheme is established under the [Carbon Credits \(Carbon Farming Initiative\) Act 2011](#). Participation in the ACCU Scheme is voluntary.

ACCUs

An ACCU is a regulated, tradeable financial unit. One ACCU represents one tonne of CO₂-e stored or avoided by an ACCU project.

The Australian Government's Clean Energy Regulator administers the ACCU Scheme. It issues ACCUs for an approved project once the project proponent shows their project has reduced emissions or stored carbon and submits an offsets report. Before issuing ACCUs, the Clean Energy Regulator verifies the reported emissions reductions or carbon stored against the eligibility requirements of the applicable method. Then, the Clean Energy Regulator issues ACCUs in the [Unit and Certificate Registry](#), under the proponent's account in the [Australian National Registry of Emissions Units](#) (ANREU) system. The proponent can sell or keep their ACCUs.

The Clean Energy Regulator's [ACCU Scheme project register](#) has a summary of registered ACCU projects and ACCUs issued and relinquished (returned).

ACCU Scheme project eligibility requirements

An ACCU Scheme project must:

- follow a legislated method
- be new: the activity cannot have started before being registered with the Clean Energy Regulator
- not be required to be conducted by Commonwealth, State or Territory law or regulation
- not be already funded by a government program listed in the Carbon Credits (Carbon Farming Initiative) Rule 2015
- not be a type of project defined as an excluded offsets project in the Carbon Credits (Carbon Farming Initiative) Rule 2015.

Some ACCU Scheme methods provide exceptions to the requirements to be a new activity or not being required by regulations.

These requirements aim to ensure projects deliver genuine emission reductions or carbon storage that would not have otherwise occurred.

You can review [eligibility requirements](#) and use the Clean Energy Regulator's interactive Australian Carbon Credit Unit Scheme Questionnaire to help determine if your project meets the ACCU Scheme's eligibility requirements.

Other requirements not covered in this topic may also apply to your project. More information about [how to participate](#) in the scheme is available from the Clean Energy Regulator. Section 3 outlines the steps in establishing a project.

ACCU Scheme methods

An ACCU Scheme project must follow a specified method to be eligible to earn ACCUs. Methods spell out rules for conducting ACCU Scheme projects, including:

- eligible project activities
- requirements about conducting project activities
- rules for estimating emission reductions and carbon storage
- monitoring, reporting and record-keeping requirements.

Methods are legislative instruments approved by the Australian Government. They provide for projects delivering real emissions reductions and carbon storage that can count towards Australia's emissions reduction commitments. Methods must meet legislated standards. The standards include additionality, which means reducing or avoiding emissions or storing more carbon that would be unlikely to occur in the ordinary course of events.

Approved methods and supporting materials are available at the Clean Energy Regulator's website [ACCU Scheme methods](#).

The number of methods changes as existing methods are retired and new methods are introduced. When a method is closed, projects already registered under that method can continue. New projects may not be registered under closed.

In section 6, there is an activity for you to explore the method that most interests you.

ACCU Scheme permanence obligations and risk of reversal buffer

Because of human-induced or natural events (such as bushfires), the carbon an ACCU Scheme vegetation or soil carbon project stores can be released back into the atmosphere. Therefore, projects that store carbon have [permanence obligations](#). The scheme considers carbon storage to be permanent if the carbon is stored for 100 years.

A permanence obligation is a requirement to maintain the carbon stored for a set minimum period of time, including preparing a permanence plan and taking steps to deal with any loss of stored carbon. These steps may involve restoring lost carbon stores or relinquishing ACCUs already issued. The permanence obligations webpage provides more details. Farmers and land managers should closely consider the implications of permanence obligations and permanence risks for their plans.

When registering a project with a permanence obligation, a project proponent must choose a permanence period of either 25 or 100 years. They can't change this period after they nominate it.

A project proponent that chooses a 25-year permanence period is issued fewer ACCUs than if they had chosen a 100-year period. This is called a permanence period discount. It helps manage the scheme-wide risk to the Australian Government if carbon stores are not maintained after projects with 25-year permanence periods conclude. For most methods, proponents choosing a 25-year permanence period receive 20% fewer ACCUs.

This permanence period discount is additional to the 5% [risk of reversal buffer](#), which applies to all projects that store carbon. The risk of reversal buffer reduces the ACCUs issued during a reporting period (see section 4) by 5%.

If a property where there is a project with permanence obligations is sold, the obligations continue to apply for the duration of the permanence period.

Source: <https://www.agriculture.gov.au/agriculture-land/farm-food-drought/climatechange/carbon-farming-outreach-program/training-package/topic-5/2-australian-carbon-credit-units>